

Chapter 1

Financial Markets

1.1 Introduction to financial markets

Q. Which markets are efficient?

- The **Weak-Form EMH** asserts that stock prices already reflect all information that can be delivered by examining market trading data (e.g., past prices, trading volumes, interest rates)
- The **Semistrong-Form EMH** is associated with all publicly available information (e.g., accounting data)
- The **Strong-Form EMH** is associated with all information relevant to the firm (e.g., insiders's information)

◇ Real Assets VS Financial Assets

- real assets: relevant to productivity directly. (e.g., plants, machines, patents)
- financial assets: claims on real assets.

◇ In economics, a financial market is a mechanism that allows people to buy and sell (trade) financial securities (e.g., stocks, bonds), and other fungible items of value at low transaction costs and at prices that reflect the Efficient Market Hypothesis (EMH). In finance, financial markets facilitate:

- the raising of capital (in the capital markets)
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- the transfer of risk (in the derivative markets)
- international trade (in the currency markets)

and are used to match those who *want* capital to those who *have* it.

◇ The financial markets can be divided into different subtypes:

- capital markets which consist of:
 - stock markets, which provide financing through the issuance of shares or common stock, and enable the subsequent trading thereof.
 - bond markets, which provide financing through the issuance of bonds, and enable the subsequent trading thereof.
- money markets, which provide short term debt financing and investment.
- derivatives markets, which provide instruments for the management of financial risk.
- foreign exchange markets, which facilitate the trading of foreign exchange.

1.2 Options

1.2.1 Call option VS put option

◇ The simplest financial option, a European call option, is a contract with the following conditions:

- at a prescribed time in the future, known as the **expiry (date)** or **expiration (date)**, the holder of the option may
- purchase a prescribed asset, known as the **underlying (asset)** for a
- prescribed amount, known as the **exercise price** or **strike price**.

◇ The option to buy an asset discussed above is known as a **call option**. The right to sell an asset is known as a **put option** and has payoff properties which are opposite to those of a call.

Q. (financial problems for options)

- How much would one pay for this right? What is the value of an option?
- How can the writer minimize the risk associated with his obligation?

1.2.2 European type VS American type options

◇ An **American option** is an option that may be exercised at *any time* prior to expiry. The options described above, which may only be exercised *at* expiry, are called **European options**. To the mathematician, American options are more interesting since they can be interpreted as free boundary problems. Not only must a value be assigned to the option but, and this is a feature of American options only, we must determine *when it is best to exercise the option*.

1.2.3 Path-dependent VS path-independent options

◇ Other types of option include the so-called **path-dependent** options. These options have values which depend on the history of an asset price, not just on its value on exercise. There are three well-known path-dependent options:

- barrier options (the option can either come into existence or become worthless if the underlying asset reaches some prescribed value before expiry)
- Asian options (the price depends on some form of average)
- lookback options (the price depends on the asset price maximum or minimum)

1.3 Other financial products

1.3.1 Bonds

◇ A **bond** is a contract, paid for up-front, that yields a known amount on a known date in the future, the **maturity (date)**.

- face value or principal
 - The bond may also pay known cash dividends (**coupons**) at fixed times during the life of the contract.
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- If there is no coupon the bond is known as a **zero-coupon bond**. Bonds may be issued by both governments or companies.
- The main purpose of a bond issue is the raising of capital, and the up-front premium can be thought of as a loan to the government or the company.

Q. What is the main difference between normal financial products and financial derivatives?

1.3.2 Forward contracts VS futures

◇ A **forward contract** is an agreement between two parties whereby one contracts to buy a specified asset from the other,

- for a specified price, known as the **forward price**,
- on a forward price, on a specified date in the future, the **delivery date** or **maturity date**.

◇ This contract has similarities to an option contract if we think of the forward price as equivalent to the exercise price. However, forward contracts are different from option contracts in that

- the asset *has to* be delivered and paid for
- no money changes hands until delivery, whereas the premium for an option is paid up-front.

◇ A **futures (contract)** is in essence a forward contract, but with some technical modifications.

Forward contracts	Futures (Contracts)
Contracts may be set up between any two parties.	Contracts are usually traded on an exchange which specifies certain standard features of the contract such as delivery date and contract size.
No margin is required.	There is a margin requirement, a system designed to protect both parties to a futures contract against default.
The profit or loss from a contract is only realized at the expiry date.	The value of a contract is evaluated every day, and the change in value is paid to one party by the other.

1.3.3 Interest rate derivatives: IRS

◇ An **interest rate derivative** is a derivative is the right to pay or receive a notional amount of money at a given interest rate. These structures are popular for investors with customized cashflow needs or specific views on the interest rate movements (such as volatility movements or simple directional movements) and are therefore usually traded over-the-counter (OTC) markets.

- Interest rate swaps (IRSs)

1.4 Types of traders

◇ Three broad categories of traders can be identified:

- hedgers, who use derivatives to reduce the risk that they face from potential future movements in a market variable,
- speculators, who use them to bet on the future direction of a market variable,
- arbitrageurs, who take offsetting positions in two or more instruments to lock in a profit (or pursue arbitrages).

1.5 Interest rates and present value

◇ For valuing options the most important concept concerning interest rates is that of **present value** or **discounting**.

Q. How much would I pay now to receive a guaranteed amount E at the future time T ?

◇ Continuous compounding formula: Assume that interest rates are constant, r . With a constant interest rate r , money in the bank $M(t)$ grows exponentially according to an ordinary differential equation (ODE)

$$\frac{dM(t)}{M(t)} = r dt.$$

The solution of this is simply

$$M(t) = C e^{rt},$$

where C is the constant of integration. Since $M(T) = E$,

$$M(t) = E e^{-r(T-t)}.$$

- If the interest rate is a deterministic function of time, that is, $r(\cdot)$,

$$M(t) = E \exp \left(- \int_t^T r(\xi) d\xi \right).$$

- annual compounding, semiannual compounding, compounded m times per annum